

File Inclusion in PHP

PHP provides multiple ways to include external files into your script.

require — include file, stop if missing

Includes and evaluates a specified file.

If the file does **not exist or has errors**, PHP **stops execution**.

Example:

```
require 'config.php';
```

Explanation:

- Use require when the file is **essential** for your application to run.

require_once — include file only once

Includes a file **only once**, even if called multiple times.

Stops execution if file is missing.

Example:

```
require_once 'config.php';  
require_once 'config.php'; // Won't include again
```

Explanation:

- Prevents **re-declaration errors** (e.g., functions or classes).
- Same fatal behavior as require.

include — include file, continue if missing

Includes and evaluates a specified file.

If the file is **missing**, PHP **issues a warning but continues execution**.

Example:

```
include 'header.php';
```

Explanation:

- Use include for **optional files**.
- Script continues even if the file doesn't exist.

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## include\_once — include only once, continue if missing

Includes a file **once** and continues if the file is missing.

Example:

```
include_once 'header.php';  
include_once 'header.php'; // Won't include again
```

Explanation:

- Combines the benefits of include and avoiding duplicates.

## JSON File Handling in PHP

JSON (JavaScript Object Notation) is often used for configuration or data storage.

### *1-> Reading a JSON file*

```
$jsonData = file_get_contents('data.json');  
// Reads file content as string
```

```
$data = json_decode($jsonData, true);  
// Converts JSON string to PHP array
```

Result:

- \$data is now an **associative array**.

Explanation:

- `file_get_contents()` reads the file.
- `json_decode()` converts JSON to PHP.
- The second parameter `true` ensures you get an **array**, not an object.

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2-> Writing to a JSON file

```
$data = ['name' => 'Taha', 'age' => 20];  
$jsonData = json_encode($data, JSON_PRETTY_PRINT);  
// Converts array to JSON string  
file_put_contents('data.json', $jsonData);  
// Writes to file
```

Explanation:

- `json_encode()` converts PHP array to JSON string.

- `JSON_PRETTY_PRINT` makes the file readable.

```
from this: {"name":"Taha","age":20}  
to this:  
{  
    "name": "Taha",  
    "age": 20  
}
```

- `file_put_contents()` writes the string to the file (creates it if missing).